

Large Language Models (LLMs)

Large Language Models (LLMs) are advanced artificial intelligence systems trained on vast amounts of text data. They use deep learning architectures, primarily Transformers, to understand and generate human-like language.

Key Characteristics

- Trained on massive text corpora
- Capable of text generation, summarization, translation, and question answering
- Learn statistical patterns in language rather than explicit rules
- Can be adapted to specific tasks through prompting or fine-tuning

Common Applications

Chatbots, virtual assistants, content creation, code generation, document analysis, research support, customer service automation, and educational tools.

Examples

GPT, Claude, Gemini, Llama, Qwen, and Mistral are examples of modern LLM families.

Challenges

LLMs may generate incorrect information (hallucinations), inherit biases from training data, and require significant computational resources for training and deployment.